

Data Stream IV: CountMin Sketch Algorithm

COMPSCI 753

The University of Auckland

RECAP: Misra-Gries Summary ('82)

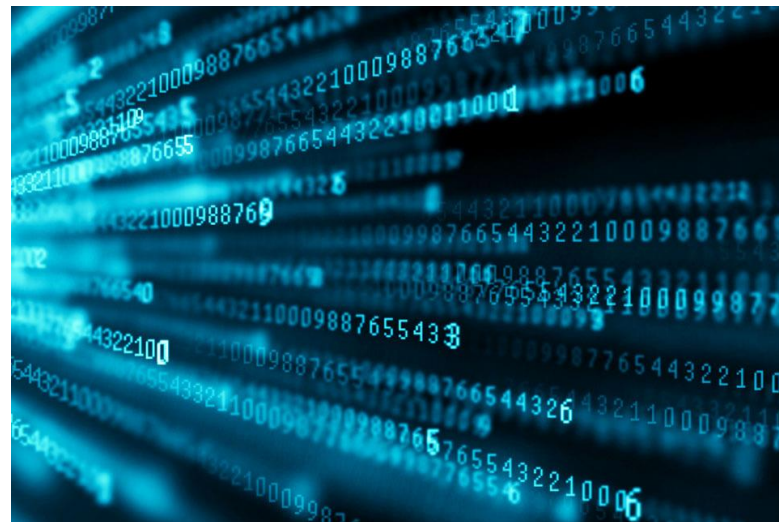
- Deterministic Solution: maintain a list **L** of **k** pairs of the form (key, value), **1 pass**
- Given an update to item *i*
 - If: *i* is in **L**, increment its counter by 1
 - Else If: *i* is not in **L** and there are fewer than **k** pairs in **L**, create its counter and initialized to 1
 - Else: subtract 1 from all counters in **L**. Remove pairs with value 0
- Claim: Estimate is smaller than exact count by at most $(m - m')/(k + 1)$
 - We can find the most frequent items if their frequencies $> (m - m')/(k + 1)$

$a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}$
{**1**, **2**, 3, **1**, 4, **2**, **1**, 4, 5, **2**, **6**}, **k**=3, **m**=11

$$\frac{(m-m')}{(k+1)} = \frac{11-3}{4} = 2 \text{ steps}$$

Outline

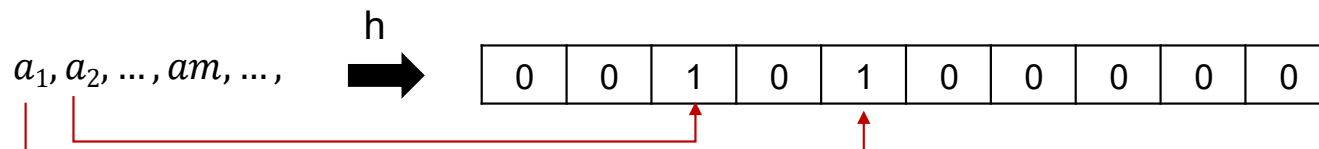
1. Sampling from a Data Stream
2. Filtering a Data Stream: Bloom Filters
3. Finding Heavy Hitters in Data Stream
 - Misra-Gries Algorithm
 - CountMin Sketch Algorithm
 - Algorithm Illustration
 - Theoretical Bound
 - Count Sketch Algorithm



Data as Infinite and Non-stationary

Counting Problem

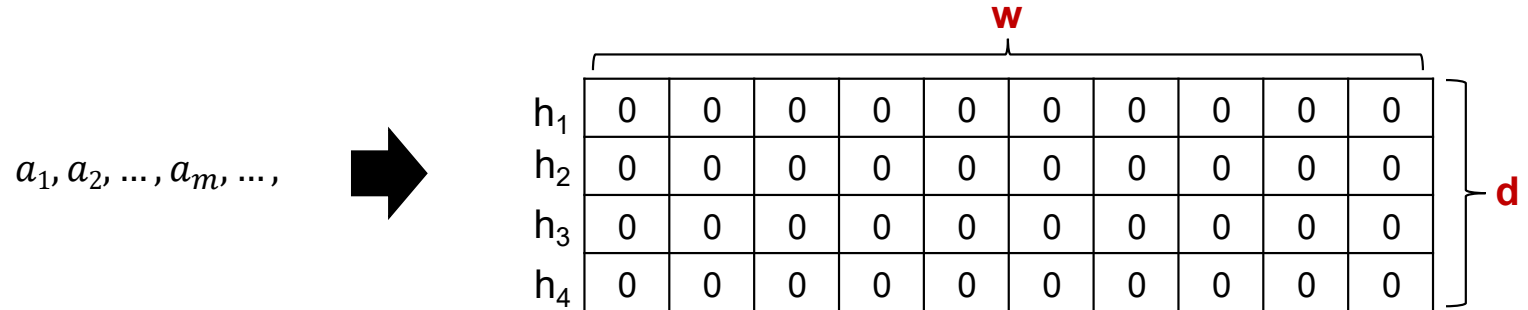
- Given an input stream $a_1, a_2, \dots, a_m, \dots$, each a_i from a universe U , return the number of times a value x appears, i.e., frequency of x (denoted by f_x)
- Example: {1, 2, 3, 1, 4, 2, 1, 4, 5, 2, 6}, $n=6$, $m=11 \rightarrow f_{x=1} = 3$
- Naïve solution:** Using universal hashing function h
 - Initially each element set to zero
 - Hash each a_i to a bucket and increase the counter



- Is there any issue for this method?

CountMin Sketch

- CountMin sketch data structure: matrix of counters
 - Initially each element set to zero
 - Hash each element in the stream with d hash functions



Example: Insertion, Update

$h_1(a_1)$	h_1	0	0	0	0	0	0	0	0	0
$h_2(a_1)$	h_2	0	0	0	0	0	0	0	0	0
$h_3(a_1)$	h_3	0	0	0	0	0	0	0	0	0
$h_4(a_1)$	h_4	0	0	0	0	0	0	0	0	0

Red arrows indicate updates: h_1 at index 5, h_2 at index 4, h_3 at index 7, and h_4 at index 10. Each update is marked with a red $+1$.

h_1	0	0	0	0	1	0	0	0	0	0
h_2	0	0	0	1	0	0	0	0	0	0
h_3	0	0	0	0	0	0	0	0	0	0
h_4	0	0	0	0	0	0	0	0	0	1

Red arrows indicate updates: h_1 at index 3, h_2 at index 2, h_3 at index 7, and h_4 at index 4. Each update is marked with a red $+1$.

Collision: $h_3(a_1) = h_3(a_2)$

Example: Query

- **Query:** find our estimation of $\tilde{f}_{x=q_1}$ using CountMin sketch matrix counter
- How many functions suggest the collisions with q_1 ?
- 2 functions

h_1	0	0	1	0	1	0	0	0	0	0	q_1 never appears
h_2	0	1	0	1	0	0	0	0	0	0	definitely collide with 1 item
h_3	0	0	0	0	0	0	2	0	0	0	definitely collide with 2 items
h_4	0	0	0	0	1	0	0	0	0	1	q_1 never appears

- **Answer:** $\tilde{f}_{x=q_1} = 0$ because q_1 never appears in the data stream we have seen so far

Example: Query

- **Query:** find our estimation of $\tilde{f}_{x=q_2}$ using CountMin sketch matrix counter
- How many functions suggest the collisions with q_2 ?
- At least 1 function

h_1	0	0	1	0	1	0	0	0	0	0	might collide with 1 item
h_2	0	1	0	1	0	0	0	0	0	0	might collide with 1 item
$h_{\{1,2,3,4\}}(q_2)$	0	0	0	0	0	0	2	0	0	0	definitely collide with 2 items
h_4	0	0	0	0	1	0	0	0	0	1	might collide with 1 item

- **Answer:** $\tilde{f}_{x=q_2} = 1$, q_2 might appear in the data stream or it simply collide with items
 $\uparrow$
 $\min(1,1,2,1)$

Example: Heavy Hitters

- **Query:** find our estimation of $\tilde{f}_x$ using CountMin sketch matrix counter
- **Answer:** $\tilde{f}_x = \min(M_{1,h_1(x)}, M_{2,h_2(x)}, M_{3,h_3(x)}, M_{4,h_4(x)}) = 15$

$h_{\{1,2,3,4\}}(x)$	h_1	15	22	20	1	13	46	32	25	18	16
	h_2	10	16	14	13	30	19	45	23	7	31
	h_3	16	1	31	17	13	18	23	45	23	21
	h_4	17	22	3	31	6	15	54	6	25	29

Red boxes highlight the values 20, 30, 23, and 15. Red '+1' markers are placed above the 20, 46, 30, 19, 23, and 15 cells.

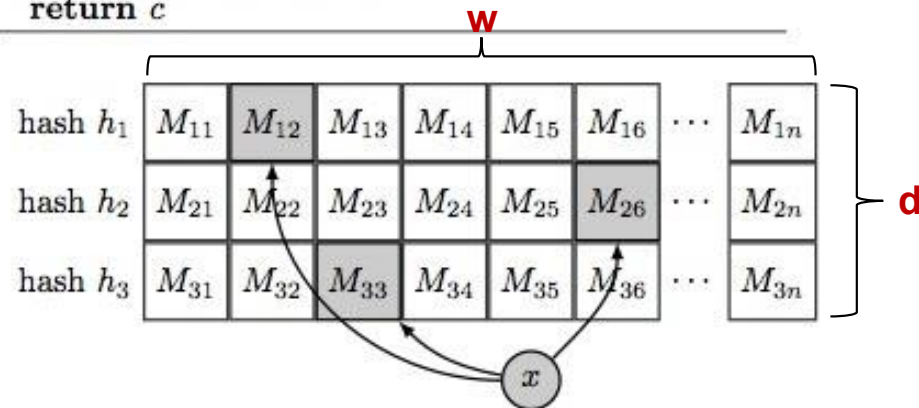
- **Query:** given frequency threshold 20, is x a potential heavy hitter?
- **Answer:** No

Randomized: CountMin Sketch

- Given
 - d independent **universal** hash functions h over range $[0, w)$
 - d different counters, $M_1, \dots, M_d$.
 - Each of size w initialized with 0s.
- Update to item x :
 - For each hash function, compute $h_i(x)$ and increase $M_{i,h_i(x)}$ by 1
- Query x
 - Return $\tilde{f}_x = \min(M_{1,h_1(x)}, \dots, M_{d,h_d(x)})$

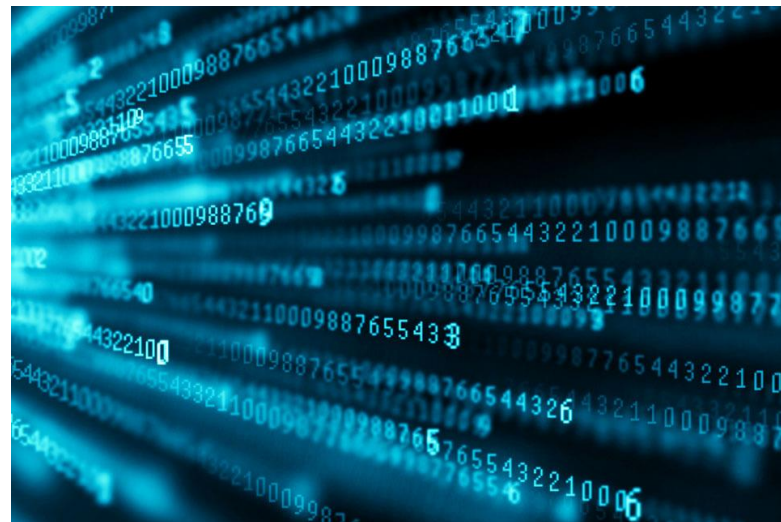
Algorithm 1 Count-Min Sketch

```
insert( $x$ ):  
  for  $i = 1$  to  $d$  do  
     $M[i, h_i(x)] \leftarrow M[i, h_i(x)] + 1$   
  end for  
  
query( $x$ ):  
   $c = \min \{M[i, h_i(x)] \text{ for all } 1 \leq i \leq d\}$   
  return  $c$ 
```



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 - **Theoretical Bound**
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Data as Infinite and Non-stationary

RECAP: Universal Hash Function Family

- Universal hash function definition:
 - A family of hash function $H = \{h : U \rightarrow \{0, 1, \dots, w - 1\}\}$ is universal if for any 2 distinct keys $x_i \neq x_j \in U$, the following holds:

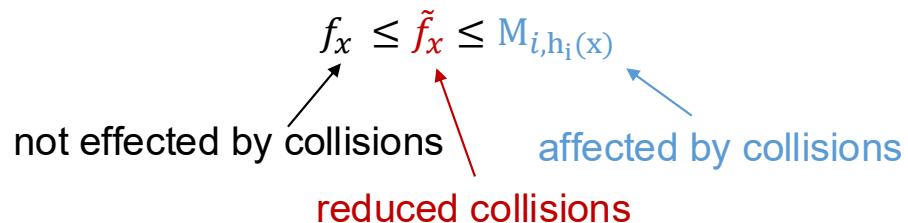
$$Pr_h[h(x_i) = h(x_j)] \leq 1/w$$

- CountMin sketch (CM):
 - Given two different items $x_i \neq x_j$, what is the probability that x_i and x_j collide?

$$Pr_h[h(x_i) = h(x_j)] \leq 1/w$$

CountMin Sketch Observations

- The relationship between the following three measures
 - True frequency: f_x
 - Estimated frequency: $\tilde{f}_x = \min(M_{1,h_1(x)}, \dots, M_{d,h_d(x)})$
 - Single estimated frequency: $M_{i,h_i(x)}$



- Collisions results in overestimation of frequency

Diagram illustrating a CountMin sketch matrix with width w and depth d .

15	22	20	1	13	46	32	25	18	16
10	16	14	13	30	19	45	23	7	31
16	1	31	17	13	18	23	45	23	21
17	22	3	31	6	15	54	6	25	29

Red boxes highlight the values 20, 30, 23, and 15. Red arrows from these boxes point to a circle labeled "min", indicating the minimum value used for the estimated frequency.

Theoretical Claim

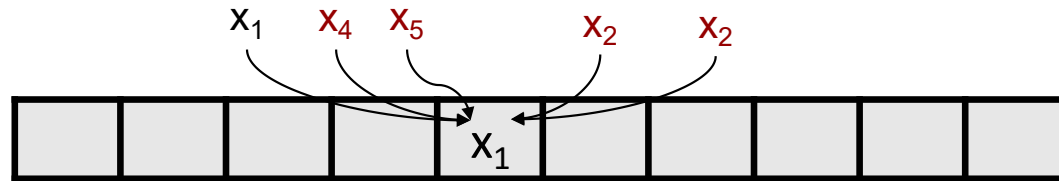
- Parameter Setting: $w = 2/\varepsilon$ and $d = \log\left(\frac{1}{\delta}\right)$, where
 - ε is the error tolerance
 - m is the number of items in the stream
 - δ is the probability to fail
- CountMin Property: Probability that estimated frequency is within a tolerable error (εm) approaches to 1

w									
15	22	20	1	13	46	32	25	18	16
10	16	14	13	30	19	45	23	7	31
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Phase 1: tolerable error
Phase 2: successful probability

Algorithmic Analysis: 1 Hash Function

- **1 hash function h:** Given a particular item x_1 , how many times $x_i \neq x_1$ collide by h?



x_4 contributes once
 x_5 contributes once
 x_2 contributes twice

- Proof :
 - Let $c_2, \dots, c_n$ be contributions of $x_2, \dots, x_n$ in the bucket $h(x_1)$
 - $\Pr [h(x_i) = h(x_1)] \leq 1/w$ ← Universality
 - Frequency increments by item x_i in expectation $E [c_i] \leq f_i / w$ ← Reoccurring items like x_2
 - Let $Z = c_2 + \dots + c_n$ be the total contributions by other item $x_i \neq x_1$ ← Over-count due to collisions, i.e., error
 - $E[Z] = \sum_{i=2}^n E[c_n]$ ← Linearity of expectation

$$= \sum_{i=2}^n \frac{f_i}{w} = \frac{\sum_{i=2}^n f_i}{w} \leq \frac{\sum_{i=1}^n f_i}{w} = \frac{m}{w}$$

← Per hash function over-count is at most $\frac{m}{w}$

Algorithmic Analysis: 1 Hash Function

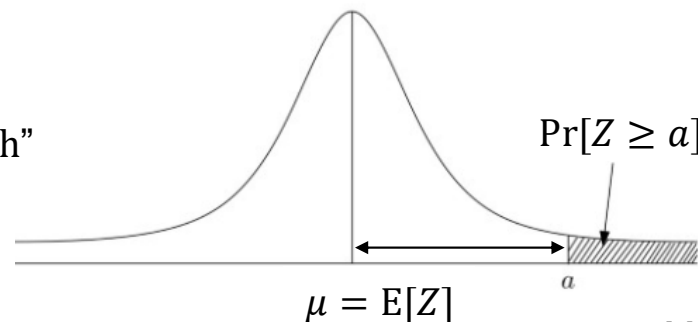
- Using Markov's inequality

- Let Z be the “total increments by other item $x_i \neq x_1$ by h ”

- $\Pr[Z \geq a] \leq \frac{E[Z]}{a}$ $\leftarrow$ Markov's inequality

- $\Pr[Z \geq \epsilon m] \leq \frac{E[Z]}{\epsilon m}$ $\leftarrow$ Let a be ϵm

- $\Pr[Z \geq \epsilon m] \leq \frac{1}{\epsilon w}$ $\leftarrow E[Z] \leq \frac{m}{w}$



Prob. of the shaded region is bounded, i.e., $\Pr[Z \geq a] \leq \frac{E[Z]}{a}$

- Proof: Successful Probability

- $$\begin{aligned} \Pr[\tilde{f}_1 \geq f_1 + \epsilon m] &= \Pr[f_1 + Z \geq f_1 + \epsilon m] \leftarrow \tilde{f}_1 = f_1 + Z, \text{ CountMin always overestimate by at most } Z \\ &= \Pr[Z \geq \epsilon m] \\ &\leq \frac{1}{\epsilon w} \\ &\leq 1/2 \end{aligned}$$
 $\leftarrow$ Let $w = 2/\epsilon$, prob. our error is larger than ϵm is bound at most 1/2 (Good)

Algorithmic Analysis: d Hash Functions

- **d hash functions $h_1, \dots, h_d$:** Given a particular item x_1 , the estimated frequency is as follows:

$$\tilde{f}_1 = \min(M_{1,h_1(x_1)}, \dots, M_{d,h_d(x_1)}) = \min(f_1 + Z_1, f_1 + Z_2, \dots, f_1 + Z_d) \leq f_1 + \epsilon m$$

- **Boosting the accuracy:** Using **d** independent hash functions corresponding to **d** independent arrays of counters

- $\Pr[\forall_{1 \leq i \leq d} Z_i \geq \epsilon m] \leq \left(\frac{1}{2}\right)^d$ $\leftarrow$ d pushes the prob. close to zero quickly
 $\leq \delta$ $\leftarrow$ Let $d = \log\left(\frac{1}{\delta}\right)$, i.e., δ to get extreme overestimate
- $\Pr[\exists_{1 \leq i \leq d} Z_i < \epsilon m] \geq 1 - \delta$ $\leftarrow$ Compliment

- With probability at least $1 - \delta$, we have a probability that our error is within ϵm (Good)

$\tilde{f}_1$ is not too far from f_1 with probability $1 - \delta$

$\uparrow$ $\uparrow$

phase 2 (ϵm) phase 2

CountMin Sketch Data Structure Size

$w = 2/\epsilon$

15	22	20	1	13	46	32	25	18	16
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$d = \log\left(\frac{1}{\delta}\right)$

- CountMin Sketch must contain $O(wd) = O(\epsilon^{-1} \log \delta^{-1})$ counters
- Example
 - $\epsilon = 10\%$ number of input data streams being overestimated, $\delta=0.1 \rightarrow wd \approx 20$ entries
 - $\epsilon = 1\%$ number of input data streams being overestimated, $\delta=0.01 \rightarrow wd \approx 400$ entries

Summary of CountMin Sketch

- Randomization Technique for Finding Heavy Hitters
 - CountMin sketch produced overestimated frequency
 - Overestimation due to collisions
- Theoretic Bound
 - Good successful probably bound on tolerable overestimation

Resources

- **[Book]** Mining of Massive Datasets by Jure Leskovec, Anand Rajaraman, and Jeff Ullman. Chapter 4
 - <http://mmds.org>
- **[Book Optional]** Machine Learning for Data Streams with Practical Examples in MOA (<https://moa.cms.waikato.ac.nz/book-html/>). Chapter 4.6.2
- Multi-Purpose Sketches
 - <https://people.cs.umass.edu/~mcgregor/711S18/vectors-3.pdf>